SAC 3 Revision Pack — Probability & Statistics

VCE Specialist Mathematics 3&4 · Chapters 15–16 · SAC in Week 7

Full solutions at the back. Work the questions before reading them.

Checklist — can you...

- ☐ compute $E(aX+b)$, $\text{Var}(aX+b)$, $\text{sd}(aX+b)$
- ☐ compute $E(aX+bY)$ and, for independent variables, $\text{Var}(aX+bY)$
- ☐ explain why variances *add* under subtraction
- ☐ distinguish $2X$ from X_1+X_2 (and nX from $\sum X_i$)
- ☐ state that linear combinations of independent normals are normal
- ☐ state the distribution of $\bar{X}$: $N(\mu,\ \sigma^2/n)$
- ☐ invoke the CLT correctly (large n, any population)
- ☐ use the normal approximation to the binomial
- ☐ build and interpret a 95% (90%, 99%) confidence interval
- ☐ find n for a given interval width
- ☐ run a full z-test: hypotheses, statistic, p-value, decision, context
- ☐ choose one-tail vs two-tail from the wording
- ☐ describe Type I and Type II errors in context; $P(\text{Type I}) = \alpha$

Questions

R1. X has mean 12 and standard deviation 3. Find the mean and standard deviation of $Y = 5 - 2X$. *(3 marks)*

R2. A and B are independent with $\text{Var}(A) = 9$, $\text{Var}(B) = 16$. Find $\text{Var}(3A - 2B)$. *(2 marks)*

R3. $X \sim N(250,\ 5^2)$ g. Write down the distributions of $2X$ and of X_1+X_2 (independent copies), and compute $P(2X > 510)$ and $P(X_1 + X_2 > 510)$. Explain the difference in one sentence. *(6 marks)*

R4. A latte is made from espresso $E \sim N(40,\ 2^2)$ mL and milk $M \sim N(160,\ 5^2)$ mL, independently. Find the distribution of the total volume, and the probability a latte is less than 195 mL. *(4 marks)*

R5. Scores are $N(80,\ 12^2)$. For a random sample of 9, find $P(\bar{X} > 86)$. *(3 marks)*

R6. $X \sim \text{Bi}(200,\ 0.3)$. Use a normal approximation to estimate $P(X \le 50)$. *(3 marks)*

R7. A sample of 49 calls has mean duration $\bar{x} = 6.8$ minutes; $\sigma = 1.4$ minutes. Construct a 95% confidence interval for μ and interpret it. *(4 marks)*

R8. Measurements have $\sigma = 2.5$. Find the smallest sample size for which a 95% confidence interval has width at most 1. *(3 marks)*

R9. Widget masses have historically had $\mu = 50$ g with $\sigma = 8$ g. After a supplier change, 32 widgets have $\bar{x} = 52.5$ g. Test at the 5% level whether the mean has *changed*. Show all steps. *(6 marks)*

R10. For your test in R9: describe a Type I error and a Type II error in context, and state which one you risk making given your conclusion. *(3 marks)*

Solutions

R1. $E(Y) = 5 - 2(12) = -19$; $\mathrm{sd}(Y) = |-2| \times 3 = 6$. (3)

R2. $\mathrm{Var} = 9(9) + 4(16) = 81 + 64 = 145$. Coefficients square; the minus sign dies. (2)

R3. $2X \sim N(500,\ 100)$; $X_1 + X_2 \sim N(500,\ 50)$. $P(2X > 510) = P(Z > 1) \approx 0.159$; $P(X_1 + X_2 > 510) = P\Big(Z > \frac{10}{\sqrt{50}}\Big) = P(Z > 1.414) \approx 0.079$. The independent copies partly cancel each other's variation, so the sum is less spread than the doubled single measurement. (6)

R4. $L = E + M \sim N(200,\ 29)$, $\mathrm{sd} = \sqrt{29} \approx 5.385$. $P(L < 195) = P\big(Z < \frac{-5}{5.385}\big) = P(Z < -0.928) \approx 0.177$. (4)

R5. $\bar{X} \sim N(80,\ 4^2)$ since $\mathrm{sd} = 12/\sqrt{9} = 4$. $P(\bar{X} > 86) = P(Z > 1.5) \approx 0.067$. (3)

R6. $X \dot\sim N(60,\ 42)$ $(np = 60,\ np(1-p) = 42)$. $P(X \le 50) \approx P\Big(Z \le \frac{-10}{\sqrt{42}}\Big) = P(Z \le -1.543) \approx 0.061$. (3)

R7. $6.8 \pm 1.96 \cdot \frac{1.4}{7} = 6.8 \pm 0.392 = (6.408,\ 7.192)$ minutes. Under repeated sampling, about 95% of intervals built this way would contain the true mean duration μ. (4)

R8. $2(1.96)\frac{2.5}{\sqrt{n}} \le 1 \Rightarrow \sqrt{n} \ge 9.8 \Rightarrow n \ge 96.04$: $n = 97$. (3)

R9. $H_0 : \mu = 50$, $H_1 : \mu \ne 50$ ("changed" $\Rightarrow$ two-tail). Under H_0: $\bar{X} \sim N\bigg(50,\ \Big(\frac{8}{\sqrt{32}}\Big)^2\bigg)$, se $= 1.414$. $z = \frac{52.5-50}{1.414} = 1.768$; $p = 2P(Z \ge 1.768) \approx 0.077$. $p > 0.05$: do not reject H_0 — insufficient evidence at the 5% level that the mean mass has changed. (6)

R10. Type I: concluding the supplier change altered the mean mass when it hadn't. Type II: failing to detect a genuine change in mean mass. Having *not rejected* H_0, the error we risk is Type II. (3)